CAS (Heimdall) – system functional description

Oxygen system

ScaleAQ Automation Document CAS-FD-20, rev. A3

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Abstract

This document describes how the oxygen system of the CAS closed pen at Heimdall works, and how the control system runs it. It is written for the project team, the operators, the commissioning engineers and the oxygen supplier Normex. It covers every instrument and actuator of the oxygen system, the control principle, the start and stop sequences, the interlocks, the alarms and the fault handling, as implemented in the PLC program of 2026-10-21 and as required by the Norwegian functional description [FD]. Items that are not decided are marked **OIn** and listed in Section 6; decisions are marked **DECn**.

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1 Introduction

1.1 Purpose

Fish in the closed pen get all their oxygen from the water pumped in. The intake water does not hold enough oxygen for the biomass, so pure oxygen is dissolved into a side stream on each of the three inlets, and the dissolved O_2 in the pen is controlled to a setpoint. This document is the *system functional specification* of that oxygen system in the sense of Cooling [COO, §14.3]: it describes the system as blocks (Section 2), as functions (Section 3), and states the functional (Section 4) and non-functional (Section 5) requirements.

1.2 Scope

In scope:

- the oxygen gas supply and distribution panel up to each injector (supplier Normex), as far as it is seen by the control system;
- per inlet line $n = 1, 2, 3$: the side-stream (oxy) pump with its variable speed drive, the pressure transmitters, the O_2 injector (oxygenator), the O_2 flow meter and the O_2 control valve;
- the pen O_2 measurement (Orbit 864 multisensors) and the pen O_2 controller OIC01;
- the PLC functions, alarms, interlocks and operator functions for all of the above.

Not in scope: the inlet pumps (Framo, see the inlet FD), the emergency oxygenation ([FD] §3.2.5, a separate system without connection to the control system), and the CO_2/TDGP measurement except where it acts on the oxygen system.

1.3 Sources

Table 1: Sources. Codes in brackets are used in the text.

Code	Document	Used for
[COO]	J. Cooling, *Software Engineering for Real-Time Systems*, 2020, §14.3 System functional specifications	Structure of this document
[FD]	P-15169-CAS-FD-01 Funksjonsbeskrivelse CAS rev B (Norwegian), §3.2.4 Oksygentilførsel, tables 12–15, 17	Required functions
[AL]	P-15169-CAS-FD-02 Alarmliste CAS Heimdall rev A	Required alarms
[PID]	CAS P&ID and loop diagrams (workfolder-latex), sheets CAS-PID-200, -300, -400, CAS-LAY-700, CAS-LD-802..804, CAS-PID-900/901, -950	Tags, I/O, open points
[PLC]	Heimdall PLC project (CODESYS export of 2026-10-21): `FB.OxySystem.OxyControl`, `OxyPumpATV650`, `OxySensors`, `FB.Inlet.PenOxygenControl` (OIC01 and the line shares, called by `InletSupervisor`), `FB.Components.ControllerMode`, `FB.Pen.PenControl`, `FB.RIO.InletRIO`, DUT `System.OxySystem`, `PFC200_Sim ConfigValues`	Implemented behaviour, defaults
[PD]	CAS program design (workfolder-latex), oxygen dataflow O1–O10	Software structure
[DB]	CAS design basis (workfolder-latex) and CasDesignBasis v2.3.12	Process model, tuning
[O2S]	Beregningsark dimensjonering ved bortfall av pumper eller strøm, 17.03.26	O_2 demand, saturation targets
[CT]	Chucaotech: Oppsett innløsere CT.xlsx; brochure	Injector capacity
[NX]	Flygt NX 3153 LT 3~ 411 datasheet, 06.01.2026; MiniCAS II	Side-stream pump
[HZ]	Risikovurdering – HAZID – Oksygen (copy 2026-02-04)	Hazards
[SO]	Sensor Overview CAS V1.xlsx	Sensors
[ORB]	Orbit 864 assembly manual DOC100486-04; RDO Core IC spec sheet; RDO Core Modbus interface spec	Pen O_2 sensors
[EAV]	Schneider ATV600 communication parameters EAV64332 V4.6; ATV6xx programming manual EAV64318	Drive registers, fault codes
[NMX]	Normex: revised request sketch 15.10.26; CS Instruments VA 500 datasheet	Oxygen panel

1.4 Abbreviations

AI / AO / DI	analog input / analog output / digital input	OIC	oxygen indicating controller
DRIVECOM	CiA 402 drive state machine used by the ATV	PIC	pressure indicating controller
ETA, CMD	drive status / command word (Modbus registers)	PI	proportional–integral (controller)
FD	functional description	PLC	programmable logic controller (WAGO PFC200)
FIC	flow indicating controller	RIO	remote I/O node (WAGO 750-362 coupler)
HMI	human–machine interface	SP / PV	setpoint / process value
LOX	liquid oxygen	TDGP	total dissolved gas pressure
MX	mixer: here the O_2 injector (oxygenator)	VSD	variable speed drive (ATV650)

1.5 Revision note

Rev.	Date	Change
A	2026-10-21	First issue. Written from [FD] rev B, the P&ID set and the PLC program of 2026-10-21. Reflects the choice of Normex as oxygen supplier[DEC1].
A1	2026-10-22	RIO link loss: the line's pressure and flow values freeze with good quality (fault table, OI28). Inlet side: the inlet pumps' speed now follows the shared total of LIC01 and the PLC chooses the number of inlet pumps (CAS-FD-10 rev B); a line stops when its inlet pump is stopped by the pump count or rotation and starts again when its pump is back in REMOTE. Oxygenation is switched for the whole pen (Pen Oxygen Start / Stop) and each line has a retained Enable, replacing the per-line `Command.EnableOxyControl`; valve auto / manual as pulses. Live test 2026-10-22: the O_2 valve outputs to the 750-563 (CommonRIO words 20–22) were mapped from an unused struct and stayed at 4 mA; corrected to `OxySystem[i].OxyValve` (register OX-29), retest pending. No upstream shut-off valves (OI5 closed, budget). Drive fault reset also by acknowledging the DriveError alarm (2026-10-26).
A2	2026-10-23	OIC01 operator auto/manual bumpless both ways (register OX-30): auto→manual starts the manual output at the current output, manual→auto sets the O_2 setpoint to the PV and restarts with a cleared integral. OIC01 moved from `InletSupervisor` to its own FB `PenOxygenControl` (same behaviour); the pen state reaches it one cycle after the operator's command.
A3	2026-11-03	Drive link loss also detected when the master still reports connected but no new response has come for 2 s (freshness watchdog); the master is restarted. A communication-loss fault (ETHF / CNF) is reset automatically once the link is back, so the pump restarts in auto; the alarms stay unacknowledged until acknowledged. The reset by acknowledging DriveError (A1) was never in the program: removed from the text (register OX-31). OIC01 limit / overflow alarms only in auto.

Decisions referred to in this document. [DEC1] 2026-10-21: Normex is the supplier of the oxygen supply and distribution panel (replaces the Linde SOLVOX cabinet drawn on CAS-PID-200). [DEC2] The side-stream pump drive is a Schneider ATV650D15N4 in drive cabinet DC*n* (drawing 874.004), controlled over Modbus TCP on its embedded Ethernet port. [PD] and [DB] still name an ATV320 in places; the PLC block is `OxyPumpATV650`.

2 Block diagram description

Cooling's block diagram description [COO §14.3.2] lists the physical parts of the system and how they are connected. Figure 1 shows the oxygen system as blocks and network links; Figure 2 shows one inlet line as a simplified process diagram with the P&ID tags.

2.1 Process in one paragraph

Each of the three Framo inlet pumps (PM-31..33) pushes sea water into the pen through its inlet line. A small side stream (about 300 m^3/h, 2–4 % of the inlet flow [DB]) is taken from each inlet line by a side-stream pump (oxy pump PU4n), pressurised, and led through a Chucaotech O_2 injector (MX-4n) where pure oxygen gas is added and dissolved. The oxygen-rich side stream returns to the same inlet line and enters the pen with the main flow. The control system measures the dissolved O_2 in the pen and sets how much oxygen gas each line gets. A line can only add oxygen while its inlet pump runs: an injector on a stopped inlet cannot bring oxygen into the pen [DB].

2.2 Blocks

Oxygen supply and distribution panel (Normex)[DEC1]**.** Oxygen gas comes from a central supply (type TBD: LOX tank, generator, or the customer's supply line[OI1]) to a distribution panel. According to the Normex sketch [NMX] the panel has a DN50 header with manometer PI-01.4 (−1..+6 bar), and per line a ball valve BV-01.n, a flow meter FM-01.n, a motorised control valve MV-01.n, a pressure transmitter PT-01.n (−1..+6 bar), a manometer PI-01.n and an outlet valve OV-02.n, with a manual bypass (OV-01.n, BV-01.n) around each line and cross-over valves BV-01.7/01.8. The pneumatic shut-off valves PV-01.n and the water-in-pipe transmitter WD-01.1 are taken out of the scope. The supply pressure is marked "2.5 bar ?" on the sketch[OI1]. In this document the flow meter and the control valve keep the CAS tags of the P&ID: FM-01.n = **FT4**n, MV-01.n = **OV4**n. The oxygen supply pressure seen by the PLC is **PT40**. The interface (signals, ranges, valve type) is not yet agreed with Normex[OI2,OI3,OI8].

Side-stream pump PU4n. Flygt NX 3153 LT, 13.5 kW, 400 V, 28 A, 1455 rpm, impeller 240 mm [NX]. Duty point at 50 Hz: 5.05 m^3/min at 10.2 m head; shut-off head 18.3 m. Leakage and motor temperature sensors are standard; a Flygt MiniCAS II relay in drive cabinet DCn gives dry contacts **XS4**n (leakage) and **TSH4**n (temperature).

Variable speed drive -FC01 in DCn. Schneider ATV650D15N4, 15 kW[DEC2]. The PLC reaches it on the drive's embedded Ethernet port with Modbus TCP. The connection is opened by the line's `InletRIO[i]` function block, which runs a second, separate `SCL.sModbusMasterTCP` master for the drive (the drive has its own IP address). Drive cabinet drawing 874.004 shows no Ethernet connection yet; an RJ45 DIN-rail socket is proposed[OI20].

Pressure transmitters. PT41..43 after the pump (= before the injector) and **PT44..46** after the injector, 4–20 mA 2-wire on the line's InletRIO 750-455. A transmitter *before* the pump is not on the P&ID; the sensor overview [SO] lists six transmitters "before and after" the injector, with make and range not yet chosen[OI17].

O_2 injector MX-4n. Chucaotech CT10 per line. Capacity [CT]: 405 l/min = 34.7 kg/h O_2 at 5 % gas/liquid ratio (v/v), 1215 l/min = 104.2 kg/h at 15 %. The PLC caps the flow of one line at `FlowMax` = 104.2 kg/h, i.e. assumes 15 %[OI7].

O_2 flow meter FT4n and control valve OV4n. Both in the Normex panel and both wired to the CommonRIO (Section 4.2). The flow meter in the Normex offer is a CS Instruments VA 500 thermal mass flow meter (4–20 mA, not isolated, and Modbus RTU; unit selectable, e.g. kg/h or Nl/min) [NMX]. The PLC expects 4–20 mA = 0–150 kg/h[OI3]. The PLC drives the valve with

4–20 mA = 0–100 % opening; [FD] describes a PWM solenoid valve fed "via 4–20 mA", the Linde cabinet had a Bürkert 2865 proportional valve behind a PWM driver, and the Normex sketch shows a motorised valve[OI2].

Pen O_2 sensors. Six Orbit 864 multisensors (In-Situ RDO Core IC: O_2 in mg/L and % saturation, temperature, conductivity/salinity, pressure). Three inside the pen (**OT01..03**, used for control) and three outside (**OT21..23**, reference) [PID CAS-PID-400]. They share one RS-485 line, Modbus RTU 9600 8N1, slave addresses 1–6, through an 8-port isolated hub -MB01 to a WAGO 750-652 serial module in the CommonRIO.

Control system. WAGO PFC200 (750-8212) PLC without local I/O. All field I/O is on WAGO 750-362 Modbus TCP couplers: the *CommonRIO* (-R1-001, PLC/RIO1 cabinet) and one *InletRIO* per inlet (-R1-100 in the PLC/RIO1, RIO2 and RIO3 cabinets). The Framo pump controllers (PCCn) are OPC UA servers; the inlet pump state and heartbeat from them give the permit for the oxygen line. The HMI reads and writes `GVLMain` through the PLC's OPC UA server.

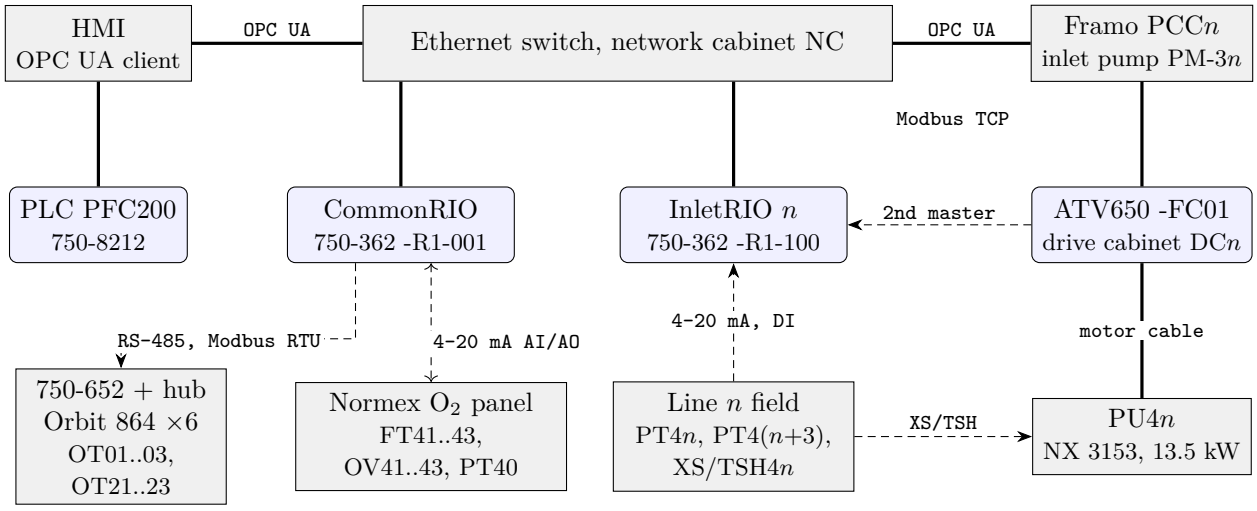


Figure 1: Block diagram of the oxygen system for one line n (three lines in total; CommonRIO, PLC, HMI and the Orbit sensors are shared). Thick: Ethernet; dashed: field signals. The drive is a Modbus TCP slave of the PLC; the "2nd master" runs in the line's `InletRIO` block.

2.3 Typical tags per line

Item	Line 1	Line 2	Line 3	PLC object
Side-stream pump, drive	PU41, DC1 -FC01	PU42, DC2 -FC01	PU43, DC3 -FC01	`OxySystem[i].OxyPump`
Pressure after pump / after injector	PT41 / PT44	PT42 / PT45	PT43 / PT46	`.Input.OxyPumpPressure` / `.OxynatorOutletPressure`
Injector	MX-41	MX-42	MX-43	–
O_2 flow, valve	FT41, OV41	FT42, OV42	FT43, OV43	`.Input.OxygenFlow`, `.OxyValve`
Controllers	PIC41, FIC41	PIC42, FIC42	PIC43, FIC43	`OxyControl[i]`
RIO node	InletRIO1	InletRIO2	InletRIO3	`InletRIO[i]`, $i = n - 1$

[FD] numbers the pressure transmitters PT41/43/45 (in) and PT42/44/46 (out); the cabinet drawings and this document use PT41–43 after the pump and PT44–46 after the injector[OI27].

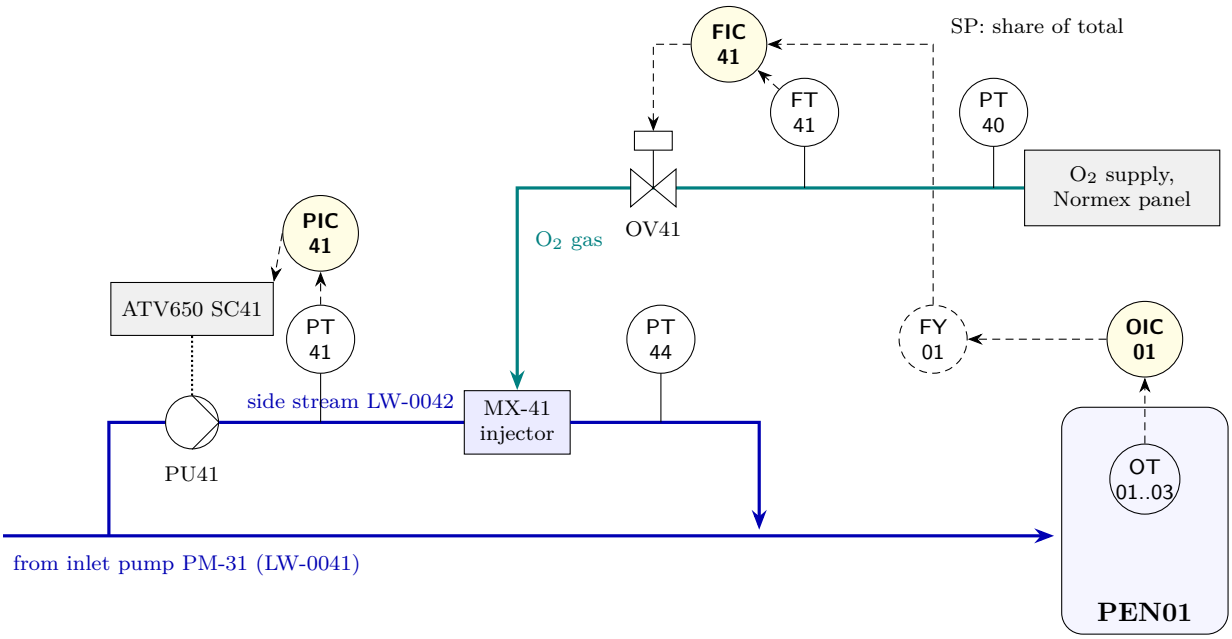


Figure 2: Simplified process diagram of oxygen line 1 with the tags of [PID CAS-PID-200/300/400] (lines 2 and 3 alike: PU42/43, PT42/43, MX-42/43, PT45/46, FT42/43, OV42/43). Blue: water; green: O_2 gas; dashed: signals. FY01 shares OIC01's total O_2 flow between the lines.

3 Functional diagram description

Cooling's functional diagram description [COO §14.3.3] shows what the system does, as functions and signal flows, independent of the hardware. The oxygen system has three layers:

1. **Pen O_2 control** (one per pen): OIC01 sets the *total* oxygen gas flow to the pen, and FY01 shares it between the lines in auto.
2. **Line flow control** (one per line): FIC4*n* sets the valve OV4*n* so the line's measured O_2 flow follows its share.
3. **Side-stream pressure control** (one per line): PIC4*n* sets the pump speed so the water pressure after the pump (into the injector) is constant.

Around these sit the line start sequence, the valve permit and the drive state machine.

3.1 Pen O_2 measurement (OY01)

The pen O_2 used for control is the mean dissolved O_2 concentration (mg/L) of the Orbit sensors that are *selected for control* (`Orbit864SensorConfig[k].EnableForAutoControl`, retained; default sensors 1–3 inside the pen), *enabled*, and of good quality. Sensors with communication failure or bad quality drop out of the mean at once. When none is left, the pen O_2 measurement is bad (`Pen.Status.OxyControlOK` = FALSE). The mean saturation (%) of the same sensors is calculated for display. [FD] asks instead for one sensor (OT03) selected by the operator, and on failure a switch to the lowest of the others; the PLC averages the selected sensors (deviation, Table 8).

3.2 Pen oxygen controller OIC01

OIC01 is a PI controller from the pen O_2 [mg/L] to the *total* oxygen gas flow to the pen [kg/h]. Because the pen responds to the total flow, its gain does not change with the number of lines running; one parameter set is enough (the three sets of [FD] table 13 are not needed).

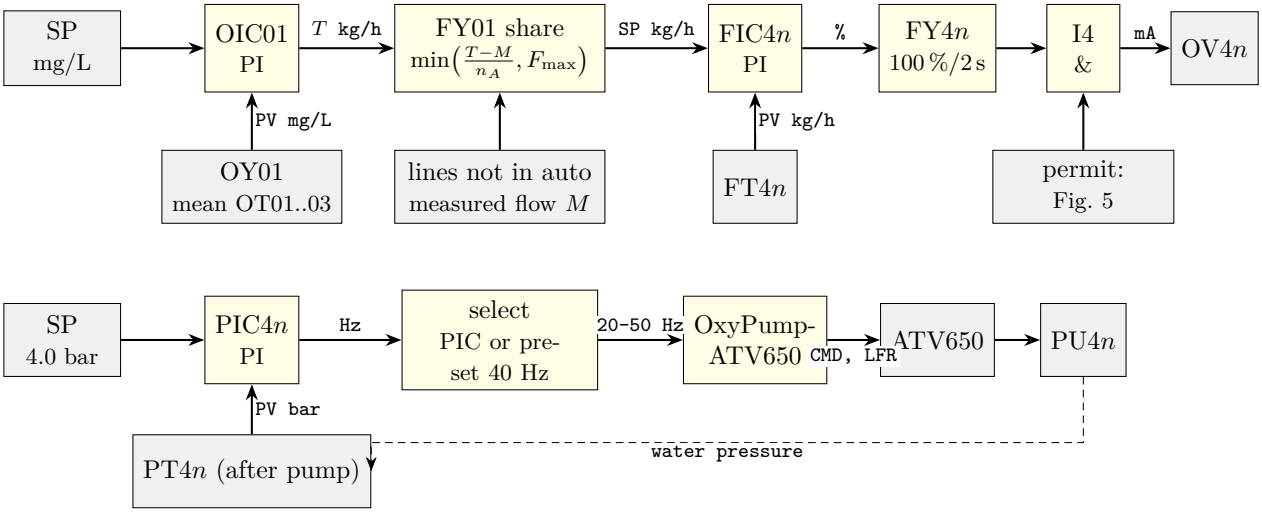


Figure 3: Control scheme [COO Fig. 7.8 style]. Top: pen O_2 cascade, OIC01 over FIC4n for each line in auto. Bottom: pressure loop of each line. T = OIC01 output, M = measured flow of the active lines not in auto, n_A = number of lines in auto, $F_{\max}$ = `FlowMax`.

- **Enabled** when the pen is started (`PenStart`) and at least one line is in auto.
- **Output limits**: 0 to the sum of `FlowMax` of the lines in auto plus the measured flow of the active lines not in auto (and `MaxOutput` if set), so the integral cannot wind up beyond what the lines can deliver.
- **Bumpless start**: on the first enabled cycle the output starts at the measured total flow.
- **Manual**: the operator can set the total flow by hand (`OxyController.Command.Manual`, `.ManualOutput`). Bumpless both ways, as LIC01 (CAS-FD-10 rev B1), while the pen is started and lines are in auto: auto→manual sets the manual output to the current total flow output, so the O_2 flow does not jump (before, it went to the last `ManualOutput`, 0 if never set); manual→auto sets the O_2 setpoint (`PenConfig.OxySetpoint`) to the pen O_2 (PV), if the measurement is valid, and restarts the PID from the manual output with a cleared integral (`RESET`, `Y\_OFFSET`). The operator then moves the setpoint to where it should be.
- **Bad O_2 measurement**: the output is frozen at its last value (interlock I7 on CAS-PID-400) for as long as the measurement is bad[OI14].

3.3 Sharing between the lines (FY01)

A line is *in auto* when oxygenation is active, the valve permit is given, the FIC is enabled, the valve is not in manual and the flow measurement is good. Every cycle:

$$\mathrm{SP}_i = \min\Big(F_{\max,i},\ \frac{T - \sum_{j \notin A} F_j}{n_A}\Big) \quad \text{for each line } i \text{ in auto } (A),$$

where F_j is the measured flow of an active line that is not in auto. A line that drops out of auto, or a manual line whose valve the operator moves, is made up by the other lines in the same cycle, not slowly through the integral of OIC01. A line not in auto shows its measured flow as its setpoint.

3.4 Line flow controller FIC4n and valve OV4n

FIC4n is a PI controller from the line's O_2 flow [kg/h] to the valve opening 0–100 %. Its output passes a rate limiter (FY4n): the valve command changes at most 100 % per `ValveRampTime` (2 s), computed with the task cycle time. The rate-limited command is written to the 750-563 as 0–65535 = 4–20 mA.

- **Auto**: SP = the line's share from FY01.
- **Valve manual**: the operator sets the opening (`OxyValve.Command.Manual`, `.ManualSetpoint`); the FIC tracks it, so the return to auto is bumpless. The line then counts as "not in auto" with its measured flow.
- **Flow measurement bad**: the FIC holds the last valve position and the line leaves auto; its flow is then counted as 0 in the sharing[OI15].
- **No permit or oxygenation not active**: output forced to 0 % = 4 mA = valve closed, also in valve manual (Section 3.7).

3.5 Side-stream pressure controller PIC4*n*

PIC4*n* is a PI controller from the pressure after the pump (PT4*n*) to the pump speed, limited to 20–50 Hz. Setpoint `PressureSetpoint` = 4.0 bar (absolute is assumed[OI17]). It is active only while the pump runs. While the pump is starting, in pump manual, with `PressureController.Enabled` = FALSE, or with a bad pressure measurement, the pump runs at the preset speed (`PresetSpeed_Hz` = 40 Hz) or, in manual, at the operator's speed. The purpose is a constant water pressure and flow through the injector, which keeps its gas transfer stable. Whether the pump can reach 4.0 bar is not confirmed: the datasheet head at the duty point is 10.2 m (about 1 bar) at 50 Hz[OI18].

3.6 Line start and stop sequence

Oxygenation is started and stopped for the whole pen (Oxygen Start / Stop, CAS-FD-10 rev B; replaces [FD]'s group start per line); each line takes part when it is enabled. Figure 4 shows the sequence.

1. The operator gives Oxygen Start (`Pen.Command.OxygenStart`, accepted while the pen is Starting or Running; `Pen.Status.OxygenOn`). Pen Stop or Oxygen Stop turns it off.
2. The line is *started* when oxygenation is on AND the line is enabled (`OxyConfig[i].Enable`, retained) AND the inlet pump of the same line is in REMOTE with a live heartbeat (`Inlet[i].InletPump.Status.OxyPermit`, interlock I5) AND no emergency stop[OI22,OI23].
3. In pump auto the oxy pump is started; it runs at the preset speed until it reaches the minimum speed, then PIC4*n* takes over.
4. When the pump has been running for `RunDelay` (30 s), *oxygenation is active*.
5. With the valve permit (Section 3.7) the valve may open and the line joins the sharing (in auto).
6. **Stop**: when the operator stops the line, the inlet permit is lost, or the pump stops, oxygenation ends at once and the valve command goes to 4 mA (closed) in the same cycle; the 2 s ramp applies only while the valve is permitted. The pump stops when the line is not started (pump in auto).

In pump manual the operator can start and stop the oxy pump alone (`OxyPump.Command.Start/Stop`), also without the inlet permit; oxygenation still needs the line to be started.

3.7 Valve permit (interlock I4)

The valve may open only when oxygen can be dissolved: the side stream must flow with pressure. `ValveOpenPressure` (3.0 bar) is on the same scale as `PressureSetpoint` (4.0 bar); if the transmitters turn out to read gauge pressure, both must be lowered together by 1 bar[OI17].

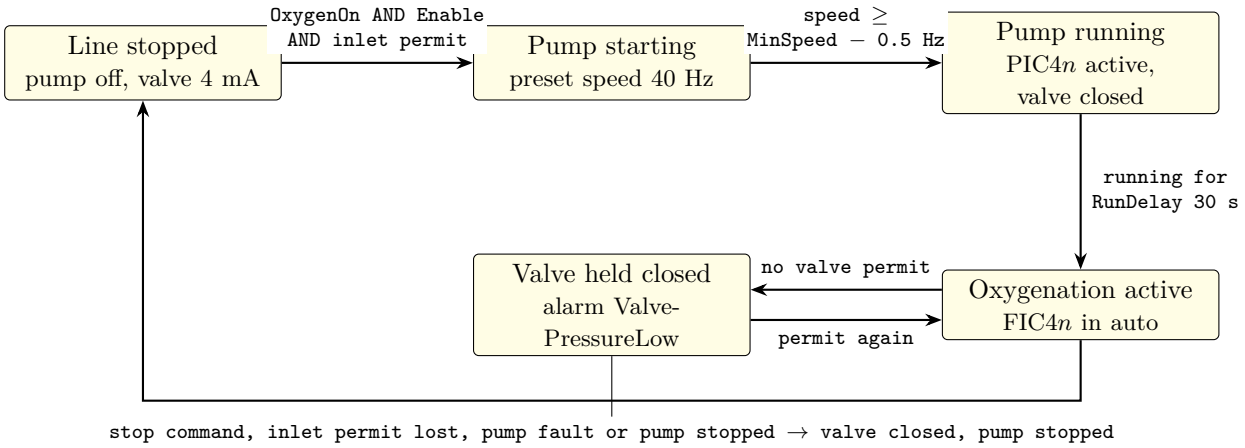


Figure 4: Line start/stop sequence (per line, pump in auto). Any state returns to "Line stopped" on the stop conditions.

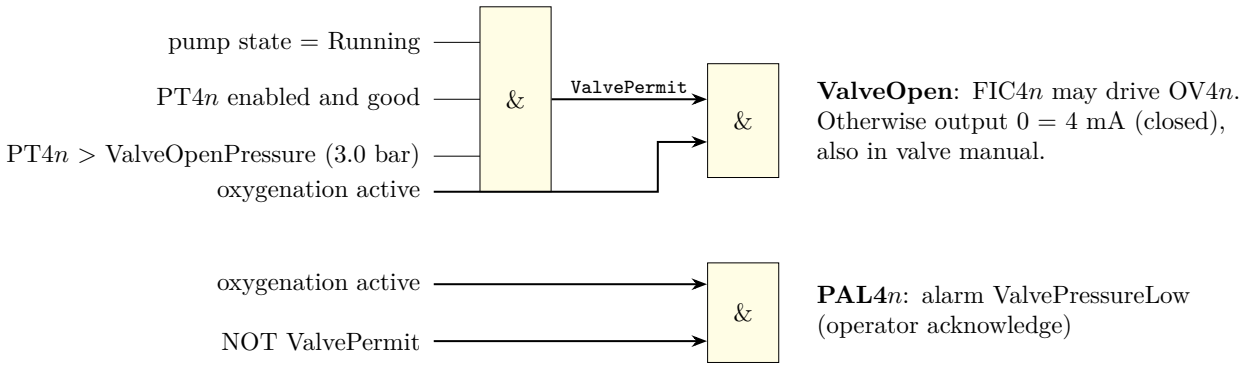


Figure 5: Valve permit I4 and alarm PAL4n (`OxyControl`).

3.8 Drive state machine (DRIVECOM)

The PLC walks the ATV650 through the DRIVECOM (CiA 402) states by writing the command word CMD (8501) and reading the status word ETA (3201) every 100 ms (block `OxyPumpATV650`). Speed reference LFR (8502) is written in 0.1 Hz, limited to 20–50 Hz. A fault is reset by the operator's Reset (`OxyPump.Command.Reset`, HMI *Reset oxy pump*), which sends `0x0080` for 0.5 s on its rising edge; a reset by acknowledging the drive fault alarm is decided but not implemented. A communication-loss fault (ETHF 150 or CNF 7, the drive's reaction to a lost link) is reset automatically: only after the PLC has itself seen the link to the drive drop and come back, once the link has been back 2 s (so the fault code LFT, read every second, is current), within 30 s and once per recovery. Any other fault, or one that the automatic reset does not clear, needs the operator. In auto the pump then starts again by itself when the line's start conditions hold; the communication and drive fault alarms stay unacknowledged until the operator acknowledges them, but do not block the restart. The link to the drive counts as lost when the master reports no connection or when no new response has come for 2 s (freshness watchdog, see the inlet FD); the master is then restarted at once and every 10 s while the data stays stale. The drive must have its command and reference channel set to the embedded Ethernet port (Fr1, Cd1).

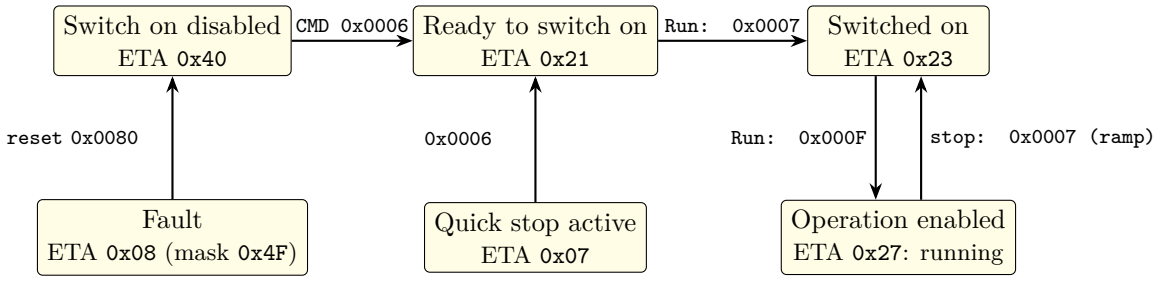


Figure 6: DRIVECOM states as handled by `OxyPumpATV650` (status masked with `0x6F`). Any other state is answered with `0x0006`. Quick stop active raises the alarm EmergencyStop.

Pump state for the logic: *Unavailable* (no communication or fault), *Available* (stopped, ready), *Starting* (Run requested, below minimum speed), *Running* (operation enabled at or above 19.5 Hz), *Stopping* (ramping down).

4 Functional requirements

This section states the functional requirements in the order of [COO §14.3.4a]. "Shall" statements are requirements; "PLC:" says how the program of 2026-10-21 fulfils them.

4.1 Processes

- F1 The system shall hold the pen O_2 at the operator setpoint by adding oxygen gas to the side stream of each running inlet (OIC01, FY01, FIC4n).
- F2 Each line shall run its side-stream pump at constant pressure into the injector (PIC4n).
- F3 A line shall add oxygen only while its inlet pump runs under remote control, its side-stream pump runs and the pressure after the pump is above the valve-open limit.
- F4 The total oxygen flow shall be shared equally between the lines in auto, up to each line's maximum; lines in manual shall be accounted for with their measured flow.
- F5 The system shall measure and display pen O_2 (mg/L, % saturation), temperature and salinity at six positions, three inside and three outside the pen.

4.2 Inputs and outputs

Table 2: Inputs and outputs of the oxygen system [PID CAS-PID-900/901].

Tag	Signal	Node / channel	PLC variable
Inputs			
PT4*n*	pressure after pump, 4–20 mA	InletRIO*n* -R1-106 ch3, word 2	`OxySystem[i].Input.OxyPumpPressure`
PT4(*n*+3)	pressure after injector, 4–20 mA	InletRIO*n* -R1-106 ch4, word 3	`OxySystem[i].Input.OxynatorOutletPressure`
XS41 / TSH41	pump leakage / motor temp., DI	InletRIO1 -R1-104 ch8/9, word 5.7/5.8 (line 1 only)	`.Input.OxyPumpLeakDetedtion` / `.OxyPumpHighTemperature`
FT41..43	O_2 flow, 4–20 mA	CommonRIO -R1-008 ch1–3, words 24–26	`OxySystem[i].Input.OxygenFlow`
PT40	O_2 supply pressure, 4–20 mA	CommonRIO -R1-008 ch4, word 27	`Pen.Input.OxygenInjectionPressure`
OT01..03, OT21..23	Orbit 864, Modbus RTU	CommonRIO -R1-003 750-652, words 0–11	`Pen.Input.Orbit864Sensor[0..5]`
SC4*n*	drive status ETA, RFR, LCR, THD, OPR, LFT	ATV650 DC*n*, Modbus TCP	`OxySystem[i].OxyPump.Input`
–	inlet permit (Framo REMOTE + heartbeat)	PCC*n*, OPC UA	`Inlet[i].InletPump.Status.OxyPermit`
Outputs			
OV41..43	valve opening, 4–20 mA	CommonRIO -R1-014 ch1/2, -R1-015 ch1, out words 20–22	`OxySystem[i].OxyValve.Output.RawPosition`
SC4*n*	CMD, LFR	ATV650 DC*n*, Modbus TCP	`OxySystem[i].OxyPump.Output`

4.3 Start-up and shut-down

S1 **PLC start** (power-up, download): all retained configuration is kept (Section 5.4); oxygenation starts off; valves at 4 mA. The operator gives Oxygen Start again. Orbit sensor selection for control is retained (default sensors 1–3).

S2 **Pen start**: OIC01 is enabled with the pen start (`Pen.Command.Start`) once a line is in auto, and starts bumplessly from the measured total flow.

S3 **Line start**: Section 3.6; [FD] requires auto restart after power return (table 2.2: auto-restart "Yes", sequence 3, after the inlet pumps). PLC: `Pen.Status.OxygenOn` is not retained over a power cut; whether oxygenation shall restart by itself is TBD[OI23].

S4 **Line stop**: operator stop, inlet permit lost or pump stopped closes the valve at once and stops the pump (in auto).

S5 **Normex supply start/stop**: local at the panel (manual valves). No PLC-controlled master shut-off valve exists in the present Normex scope[OI5].

4.4 Fault identification and recovery

Table 3: Faults, their detection and the system's response.

Fault	Detection	Response (PLC 2026-10-21)	Recovery
Drive fault	ETA state "fault"; code in LFT	Pump Unavailable, line drops out of auto, valve closed, alarms DriveError + PumpCommonError	Reset oxy pump: `0x0080`; alarm clears with a fault-free ETA; in auto the line restarts
Drive communication lost	`DriveMaster.comOk` FALSE (socket timeout 5 s), or no new response for 2 s (freshness watchdog)	Pump Unavailable, valve closed, DriveCommunicationError; the drive master is restarted at once and every 10 s. The drive's own reaction is set by ETHL (trip ETHF)	Automatic: when the link is back, an ETHF / CNF fault is reset automatically (once, after 2 s) and in auto the pump starts again; the alarms stay unacknowledged until acknowledged. [FD] wants the pump to keep running at preset speed[OI19]
Drive not in remote	ETA bit 9 = 0	Alarm DriveNotInRemote; commands ignored by the drive	Set drive to remote
Pump does not start / stop	not turning 10 s after Run; turning 10 s after stop	StartUpTimeout / StopTimeout	Operator ack
Pressure after pump low	PT4n $\leq$ 3.0 bar or bad, while oxygenation active	Valve forced closed (4 mA), alarm ValvePressureLow; PIC on preset speed if PT bad	Automatic when pressure returns; operator ack
Line flow measurement bad	750-455 diagnostic (wire break, under/over range)	Line leaves auto; valve holds last position; its flow counts as 0 in the sharing[OI15]	Automatic when good
Pen O_2 measurement lost	no selected Orbit sensor with good quality	OIC01 output frozen indefinitely[OI14]; lines keep their share	Automatic, bumpless
One Orbit sensor fails	Modbus timeout / bad quality	Sensor drops out of the mean; CommFail alarm after delay	Automatic
Inlet pump lost	inlet permit FALSE (not REMOTE or heartbeat lost)	Line stops: valve closed, oxy pump stopped; other lines take over the share at once	Automatic: while oxygenation is on and the line enabled, the line starts again when its inlet pump is back in REMOTE
Inlet pump stopped by the pump count or rotation	inlet permit FALSE	As inlet pump lost. A line whose inlet pump is started by the pump count or rotation starts when that pump reaches REMOTE, if it is enabled	Automatic while oxygenation is on[OI23]

Fault	Detection	Response (PLC 2026-10-21)	Recovery
RIO link lost (InletRIOn or CommonRIO)	Modbus TCP `CommOK` FALSE, also when no new response has come for 2 s (freshness watchdog, with automatic reconnect); alarm RIO comm. error	*None yet*: PT4n, PT4(n+3) (InletRIOn) and FT4n, PT40 (CommonRIO) keep their last value and good quality, so PIC4n, the valve permit and FIC4n run on frozen values[OI28]	–
O_2 supply lost	PT40 (displayed only)	*None*: no alarm, no interlock[OI4,OI10]; FIC opens valve fully, flow stays low	–
Water back into injector / supply	–	Mechanical only (check valves)[OI6]	–
Emergency stop	not wired[OI22]	(Drive quick stop is alarmed as EmergencyStop)	–

4.5 Communication

- **RIO**: Modbus TCP, FC23 every 100 ms; InletRIOn read 26/write 21 words, CommonRIO read 41/write 25 words (to be checked against the couplers, [PID] A1/A8).
- **Drive**: Modbus TCP, unit id 0 (TBD for the real drive[OI19]), channel 0 FC23 every 100 ms (write 8501–8502, read 3201–3211), channel 1 FC3 every 1 s (read 7121).
- **Orbit 864**: Modbus RTU 9600 8N1 over the 750-652, one sensor at a time, O_2 block every second round; stay-awake register written so the sensors do not sleep.
- **Framo**: OPC UA; heartbeat timeout 10 s.
- **HMI**: OPC UA to `GVLMain`.

5 Non-functional requirements

5.1 Control laws and start tuning

All three controllers are PI (`PIDController`, $T_d = 0$), with output limits and anti-windup. The values are start values from `ConfigValues` [DB], to be tuned at commissioning.

Loop	PV / SP	Output	K_p, T_i	Basis
OIC01	pen O_2 mg/L; SP 8.0 mg/L[OI13]	total O_2 0..$\sum$ kg/h	50 (kg/h)/(mg/L), 1800 s	Pen time constant about 44 min, gain about 0.024 mg/L per kg/h [DB]
FIC4n	O_2 flow kg/h; SP from FY01	valve 0–100 %	0.3 %/(kg/h), 5 s	Valve and meter lag about 0.2 s; closed loop about 2 s
PIC4n	PT4n bar; SP 4.0	speed 20–50 Hz	2 Hz/bar, 5 s	Pump affinity law (simulator)

Deviation from [FD] table 13/14: [FD] has OIC01 output 0–100 %, three parameter sets (P 0.5/1/2, I 600 s) by number of pumps, and FIC in l/min with P 0.5, I 600 s. The PLC works in kg/h with one set; the reason is given in Section 3 and [DB][OI16].

5.2 Timing and resolution

- PLC task `Task`: cyclic, 20 ms. Filters and the valve ramp use the task interval read from the runtime.
- RIO and drive I/O: 100 ms; drive last fault: 1 s. Orbit sensors: each sensor about every few seconds (one RS-485 line); sensor response $T_{90} < 45$ s [ORB].
- 750-455 AI: 12-bit (2048 steps over 4–20 mA): 0–10 bar → about 5 mbar, 0–150 kg/h → about 0.07 kg/h. 750-563 AO: 16-bit word.
- Drive: speed 0.1 Hz, current 0.01 A as read by the PLC (unit to be confirmed for the ATV650D15, [EAV] says "refer to programming manual"), thermal state and power 1 %.
- Orbit: O_2 0–60 mg/L, resolution 0.01 mg/L, accuracy ±0.1 mg/L (0–20 mg/L) [ORB].

5.3 Alarms

Alarms use the common `AlarmManagement` block. "auto" alarms clear themselves when the condition is gone; "ack" alarms must also be acknowledged. Drive alarms are FALSE without communication, except the communication alarm. LFT keeps the last fault after a reset, so fault codes count only while the drive is in fault.

Table 4: Alarms of the oxygen system. Codes per [EAV] (ATV600 list); "FD/AL" = required by [FD]/[AL].

Alarm (`OxySystem[i].`)	Condition	Ack	Status
Side-stream pump drive (`OxyPump.Alarm.*`), per line			
DriveCommunicationError	drive configured and Modbus TCP link not OK	auto	PLC
DriveError	ETA state fault; code in `OxyPump.Input.LastFault`	auto	PLC
DriveNotInRemote	ETA bit 9 = 0 (command from display terminal / forced local)	auto	PLC
EmergencyStop	DRIVECOM "quick stop active"	auto	PLC
DCUndervoltageError	ETA bit 4 = 0 (no mains to power part), or fault PHF (21) / USF (22)	auto	PLC
MotorTemperatureHigh	fault OLF (17), TH2F (110), TH3F (112), TH4F (120), TH5F (122)	auto	PLC
DriveTemperatureHigh	THD ≥ 100 % (`DriveThermalWarning_pct`), or fault OHF (16) / TJF (54)	auto	PLC
DriveWarning	ETA bit 7 (drive warning)	auto	PLC
PumpCommonError	any of the above except DriveWarning; [AL] "Feil på OxyPumpe n"	ack	PLC, FD/AL
StartUpTimeout	Run commanded, not turning after `OxyStartupTime` 10 s	ack	PLC
StopTimeout	stop commanded, still turning after `OxyShutdownTime` 10 s	ack	PLC
Line			
ValvePressureLow (PAL4n)	oxygenation active but no valve permit: valve held closed	ack	PLC
Sensor quality PT4n, PT4(n+3), FT4n	750-455 diagnostic bad for `QualityDelay`	ack	PLC
Low Δp over injector	PT4(n+3) – PT4n too small (limit from HMI)	–	FD/AL, not in PLC[OI10]
Pump leakage / temp.	XS4n / TSH4n	–	not in PLC[OI24]

Alarm	Condition	Ack	Status
Pen and supply			
OIC01 deviation	|PV – SP| > limit while in auto, with delay (HMI)	–	FD/AL, not in PLC[OI11]
OT*nn* low / high O_2	per sensor, limits and delay from HMI, only for active sensors	–	FD/AL; L/H in DUT, not configured[OI12]
O_2 sensor disagreement	spread between OT01..03 > limit	–	FD/AL, not in PLC[OI12]
O_2 sensor fault	Orbit communication failure (CommFail) after delay	ack	PLC
O_2 high-high shut-off	supersaturation (O_2 HH / TDGP) closes oxygenation	–	FD (HOLD), not in PLC[OI9]
PAL40 O_2 not available	PT40 low	–	FD/AL, not in PLC[OI4]

5.4 Parameters

Retained in `GVLRetain`; defaults from `PFC200_Sim` `ConfigValues` (index i = line -1).

Table 5: Configuration parameters of the oxygen system.

Parameter	Default	Meaning
`PenConfig.OxySetpoint`	8.0 mg/L	OIC01 setpoint (placeholder)[OI13]
`PenConfig.OxyController`	K_p 50, T_i 1800 s, Min 0, Max 0	OIC01; Max 0 = what the lines can deliver
`Orbit864SensorConfig[k]`	slaves 1–6; EnableForAutoControl for 1–3	sensor address, polling, selection for OIC01
`OxySystemConfig[i].OxyPump.MinSpeed_Hz/MaxSpeed_Hz`	20 / 50 Hz	PIC output range[OI19]
`.OxyPump.PresetSpeed_Hz`	40 Hz	speed while starting, with bad PT, or PIC disabled
`.OxyPump.RunDelay`	30 s	pump running before oxygen may be added
`.OxyPump.DriveThermalWarning_pct`	100 %	THD limit for DriveTemperatureHigh
`.OxyStartupTime/.OxyShutdownTime`	10 s / 10 s	start / stop timeout alarms
`.PressureController`	Enabled, K_p 2 Hz/bar, T_i 5 s	PIC4*n*
`.PressureSetpoint`	4.0 bar	PIC4*n* setpoint (abs. assumed)[OI17,OI18]
`.ValveOpenPressure`	3.0 bar	valve permit limit, same scale as setpoint
`.FlowController`	Enabled, K_p 0.3 %/(kg/h), T_i 5 s	FIC4*n*
`.FlowMax`	104.2 kg/h	most O_2 flow of one line[OI7]
`.ValveRampTime`	2 s	valve command 0–100 % at most in this time
`OxyPumpPressureConfig[i]`	0–10 bar, f_c 1 Hz	PT4*n* scaling, filter, quality

Parameter	Default	Meaning
`OxynatorOutletPressureConfig[i]`	0–10 bar, f_c 1 Hz	PT4(n+3)
`OxygenFlowConfig[i]`	0–150 kg/h, f_c 1 Hz	FT4n OI3
`InletRIOConfig[i].DriveIP/DrivePort/DriveUnitId`	sim 10.1.2.101 / 5131–5133 / 0	drive address OI19

5.5 Instruments

Table 6: Instruments and actuators of the oxygen system. TBD = not yet known.

Tag	Function	Location	Signal / interface	Range	Resolution	Use
PT40	O_2 supply pressure	Normex panel header	4–20 mA, CommonRIO -R1-008 ch4	TBD (PLC default 0–10 bar)	12 bit	display OI4
FT41..43	O_2 mass flow per line (FM-01.n)	Normex panel	4–20 mA, CommonRIO -R1-008 ch1–3	0–150 kg/h in PLC, TBD	12 bit; ±1.5 % m.v.	FIC PV OI3
OV41..43	O_2 control valve (MV-01.n)	Normex panel	4–20 mA, CommonRIO -R1-014/015	0–100 %	16 bit	FIC out OI2
PT41..43	water pressure after side-stream pump	side-stream line	4–20 mA, InletRIOn ch3	0–10 bar (PLC), TBD	12 bit	PIC PV, permit
PT44..46	water pressure after injector	side-stream line	4–20 mA, InletRIOn ch4	0–10 bar (PLC), TBD	12 bit	display
PT-01.1..3	O_2 pressure after valve	Normex panel	TBD (not connected)	−1..+6 bar	–	TBD
OT01..03	O_2, temp., salinity, inside pen	pen, near outlet	Modbus RTU, 750-652	0–60 mg/L	0.01 mg/L	OIC PV
OT21..23	O_2, temp., salinity, outside	3/30/50 m from pen	Modbus RTU, 750-652	0–60 mg/L	0.01 mg/L	reference
SC41..43	side-stream pump drive	DCn	Modbus TCP	20–50 Hz	0.1 Hz	PIC out
XS41, TSH41	pump leakage, motor temp.	DC1 MiniCAS II	DI InletRIO1 5.7/5.8	on/off	–	not used OI24
PT11, QT11	TDGP, CO_2 (Solu-Blu)	pen near outlet	4–20 mA, CommonRIO -R1-007	TBD	12 bit	display; HH? OI9

5.6 Drive Modbus registers

Checked against the ATV600 communication parameter list [EAV] (logic address = Modbus register).

Code	Reg.	Name	R/W	Unit	PLC use
CMD	8501	Command register	W	bits	DRIVECOM control word, FC23 every 100 ms
LFR	8502	Reference frequency	W	0.1 Hz	speed setpoint
ETA	3201	Status register	R	bits	state, fault, remote (bit 9), warning (bit 7), mains (bit 4)
RFR	3202	Motor frequency	R	0.1 Hz	actual speed, Running
LCR	3204	Motor current	R	see [EAV]	current (PLC: 0.01 A)
THD	3209	Drive thermal state	R	1 %	DriveTemperatureHigh
OPR	3211	Motor power in %	R	1 %	display
LFT	7121	Last error occurred	R	enum	fault code, FC3 every 1 s
ETHL	7021	Ethernet error response	cfg	enum	set in drive: factory freewheel stop; options include fallback speed LFF (7080) and speed maintain[OI19]

All register numbers and the fault codes used by `OxyPumpATV650` (16 OHF, 17 OLF, 21 PHF, 22 USF, 54 TJF, 110/112/120/122 TH2F–TH5F, communication fault ETHF 150) match [EAV]. The program design [PD] still documents the ATV320 block with codes from the ATV320 list NVE41316 (including PTFL 49 and OTFL 50, which the ATV600 list does not have); the codes shall be re-checked on the real drive's firmware at commissioning and [PD] updated[OI21].

5.7 HMI operator functions

The production HMI is not yet specified[OI25]; the present CAS HMI is a read-only overview of pen O_2, level and valve signals. The PLC offers the following operator functions in `GVLMain`; [FD] requires them on the HMI.

Table 7: Operator functions.

Function	PLC variable	[FD]
Oxygen Start / Stop (pen)	`Pen.Command.OxygenStart`, `.OxygenStop`; `Pen.Status.OxygenOn`	pulses; enabled lines follow their inlet pumps
Enable / disable oxygen line n	`OxyConfig[i].Enable`	retained
Pump auto / manual, start, stop, speed, reset	`OxySystem[i].OxyPump.Command.Auto/Manual/Start/Stop/Setpoint/Reset`	manual start allowed
Valve auto / manual, manual position	`OxySystem[i].OxyValve.Command.Manual/Auto` (pulses; mode in `.Status.Manual`), `.ManualSetpoint`	–
OIC01 setpoint	`PenConfig.OxySetpoint`	% or mg/L?[OI13]
OIC01 manual output	`OxyController.Command.Manual/ManualOutput`	–
Select sensors for control	`Orbit864SensorConfig[k].EnableForAutoControl`	selector; inactive sensors yellow
Preset speed, run delay, tuning	`OxySystemConfig[i]`, `PenConfig`	preset speed from HMI
Drive info panel	`OxyPump.Input`: ready, remote, running, power, warning, error, speed, current, LFT	table 12

Function	PLC variable	[FD]
Alarm limits (deviation, L/H, Δp)	not yet in PLC	tables 15, 17[OI11,OI12]

5.8 Deviations from the Norwegian FD

Table 8: Implemented behaviour that differs from [FD] rev B.

No.	[FD]	PLC 2026-10-21	Action
1	OIC01 output 0–100 % to the valves, or cascade to FIC as an alternative	cascade: OIC01 total kg/h shared over FIC4n	update FD[OI16]
2	three OIC parameter sets by number of oxy pumps	one set (total flow output makes the gain independent)	update FD
3	FIC SP in l/min, scaling 0–20, P 0.5, I 600 s	kg/h, 0–150, K_p 0.3, T_i 5 s	update FD; confirm unit[OI3]
4	PV = one selected sensor (OT03), switch to lowest on fault	mean of the selected good sensors	decide
5	valve: PWM solenoid “via 4–20 mA”	proportional 4–20 mA, 2 s ramp in PLC	Normex[OI2]
6	O_2 fault: freeze output (“HOLD”, sentence unfinished)	freeze indefinitely	decide[OI14]

6 Open issues and decisions

Decisions.

DEC1 2026-10-21: oxygen supply and distribution panel from Normex.
DEC2 Side-stream pump drive ATV650D15N4 on Modbus TCP (embedded Ethernet).

Open issues. Questions for Normex are marked (N).

No.	Issue	Proposed action / owner
OI1	(N) Supply type (LOX tank, generator, or customer supply line) and normal / minimum supply pressure (sketch: “2.5 bar ?”; Linde sheet: 10 bar).	Normex to state; sets PT40 range and PAL40 limit.
OI2	(N) Control valve type: [FD] PWM solenoid, PLC proportional 4–20 mA with 2 s ramp, Normex sketch motorised valve MV-01.n. Signal, stroke time, fail position (closed on signal loss?) and position feedback.	Normex; adjust `ValveRampTime` and FIC tuning to the stroke time.
OI3	(N) Flow meter unit (kg/h vs Nl/min / Nm^3/h), range for O_2, signal (4–20 mA non-isolated or Modbus RTU). PLC assumes 4–20 mA = 0–150 kg/h.	Normex; update `OxygenFlowConfig`.
OI4	(N) Low supply pressure signal: which transmitter (PT40 / PI-01.4) and limit; PAL40 “oxygen not available” and a supply pressure interlock are not in the PLC (PT40 only displayed, misnamed `OxygenInjectionPressure`).	Add PAL40 and interlock; rename.

No.	Issue	Proposed action / owner
OI5	(N) PLC-controlled master shut-off valve: *decided 2026-10-22*: no on/off solenoid or shut-off valves upstream of the control valves (budget). The control valve is the only shut-off; the PLC holds it at 4 mA without the valve permit (pump running, pressure after the pump above `ValveOpenPressure`), the inlet permit or on emergency stop. O_2 HH shut-off can only close the control valves.	Closed (register OX-17).
OI6	(N) Non-return protection: HAZID "oxygen pressure lost, water enters injector and O_2 tank". Check valve per line (Linde had CV1–3; Normex sketch unclear: OV-02.n).	Normex to confirm.
OI7	(N) Injector model and gas ratio: `FlowMax` 104.2 kg/h = CT10 at 15 % v/v; 34.7 kg/h at 5 %. Which applies in sea water at Heimdall?	Chucaotech / Normex; set `FlowMax`.
OI8	(N) Signal list and interface FD for the Normex panel (terminals, 24 V supply, cable list).	Normex.
OI9	O_2 high-high / supersaturation (TDGP) shut-off asked by [FD] (HOLD); not in the PLC. Needs limits and an actuator (OI5).	Decide limits; implement.
OI10	Injector low differential pressure alarm ([FD] table 15): `OxynatorOutletPressure` is read but unused.	Implement with HMI limit.
OI11	OIC01 deviation alarm ([FD] table 15): not implemented.	Implement.
OI12	O_2 sensor disagreement alarm and per-sensor low/high alarms: not implemented (L/H fields exist in the Orbit DUT, not configured).	Implement; set limits.
OI13	O_2 setpoint 8.0 mg/L is a placeholder; [FD] asks % or mg/L. [O2S] uses % saturation: ≥80 %, 90 % at 16 °C. A % setpoint needs temperature/salinity correction.	Customer / biology to decide.
OI14	Loss of pen O_2 measurement: OIC01 holds its output indefinitely ([FD] sentence unfinished). Hold, fixed fallback flow, or time limit?	Decide; implement.
OI15	Loss of line flow measurement: valve holds last position, line flow counted as 0, so the other lines may over-dose. Hold, close, or fixed opening?	Decide; implement.
OI16	Units and tuning differ from [FD] (Table 8); deliberate.	Accept; update [FD] to rev C.
OI17	Pressure transmitters: make, range, absolute or gauge unknown ([SO] "find sensors"). `PressureSetpoint` 4.0 and `ValveOpenPressure` 3.0 must change together. Transmitter before the pump wanted?	Select transmitters; set ranges.
OI18	Pump head vs. pressure setpoint: NX 3153 gives 10.2 m at 5.05 m^3/min, 18.3 m shut-off at 50 Hz; the simulator assumes 4.5 bar rise. Required injector inlet pressure from Chucaotech?	Check suction pressure and injector requirement; reset SP.
OI19	ATV650: IP address and unit id, comm-loss behaviour (ETHL; [FD] expects running on at preset speed), 20–50 Hz range and 40 Hz preset vs. pump curve.	Set at commissioning; decide ETHL (LFF or RLS).
OI20	Drive cabinet 874.004 has no Ethernet drawn; RJ45 DIN socket proposed.	Electrical design.
OI21	Drive fault codes: PLC uses ATV600 codes (match [EAV]); [PD] still lists ATV320 (NVE41316).	Verify on the real drive; update [PD], [DB].
OI22	Emergency stop not wired: `OxySystem[i].Input.EmergencyStop` never written; [FD] wants E-stop to stop PU4n and close OV4n.	Wire (Framo E-stop or hardwired) ([PID] A3).

No.	Issue	Proposed action / owner
OI23	Inlet permit: PLC uses inlet pump REMOTE + live heartbeat; [FD] says “inlet pump running”. Auto-restart of oxygen lines after power return ([FD] table 2.2).	Confirm REMOTE implies running; decide restart.
OI24	Pump leakage / temperature (XS, TSH) only on RIO1, not used in logic.	Add inputs on RIO2/3 and alarms ([PID] B8).
OI25	Production HMI not specified; operator functions of Table 7.	HMI design.
OI26	Oxygen capacity with one inlet pump out: two lines carry only about 560–730 t [DB]; fallback is emergency oxygenation and stopped feeding.	Project / customer.
OI27	PT numbering [FD] (PT41/43/45 in, PT42/44/46 out) vs. drawings (PT41–43 after pump, PT44–46 after injector).	Update [FD].
OI28	RIO link loss freezes the line’s analog values with good quality (register OX-28; the inlet level pairs got this fix in CAS-FD-10 rev B).	Mark the RIO’s sensors bad while `CommOK` is FALSE; line leaves auto, valve per OI15.